

STP 421 Final Review

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1 Probability Basics

1.1 Definitions and Properties of a Probability Space

Define a probability space as a triplet $(\Omega, \mathcal{F}, \mathbb{P})$ where each member of the triplet represents a property of the space. The first member of the triplet is the sample space Ω . The sample space is a set that contains all possible outcomes in the space. Any set that is a subset of the sample space is called an "Event". There aren't any notable properties of the sample space. The $\mathcal{F}$ is the sigma algebra of the space. The sigma algebra is a set that has the following properties.

- $\emptyset \in \mathcal{F}$
- $\Omega \in \mathcal{F}$
- $E \in \mathcal{F} \implies E^c \in \mathcal{F}$
- $E_1, E_2, \dots, E_n \in \mathcal{F} \implies \bigcup_{i=1}^n E_i \in \mathcal{F}$

The last member of the three musketeers is the probability measure, which is a function from the sigma algebra to the unit interval, and which has a few properties.

- $\mathbb{P}(\emptyset) = 0, \mathbb{P}(\Omega) = 1$
- $\forall_{A \in \mathcal{F}} 0 \leq \mathbb{P}(A) \leq 1$
- For a countable number of mutually exclusive events $A_1, A_2, \dots, A_n$, $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$

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1.2 Conditional Probabilities

Given two events A and B, we define the probability of A given B as follows. $\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$ where we require that $\mathbb{P}(B) > 0$ for this quotient to be defined. A note is that conditional probabilities are probability measures on a probability space, and this can be rigorously shown using elementary properties. A result of this definition is the multiplication rule, which states that $\mathbb{P}(\bigcap_{i=1}^n A_i) = \mathbb{P}(A_1)\mathbb{P}(A_2|A_1) \dots \mathbb{P}(A_n|\bigcap_{i=1}^{n-1} A_i)$.

1.3 Independence

We say that a countable collection of events is independent if for every finite subcollection of events we have that the probability of their intersection is the product of individual probabilities. Namely if we have the following events $E_1, E_2, \dots, E_n$, these events are said to be independent if for $E_{i_1}, E_{i_2}, \dots, E_{i_n}$ $\mathbb{P}(\bigcap_{r=1}^n E_{i_r}) = \prod_{r=1}^n \mathbb{P}(E_{i_r})$

¹Elementary Properties of probabilities like that of the complement, subsets, and other small details are left out to save space for more important theorems and concepts

1.4 Law of Total Probability

Let E be an event in $\mathcal{F}$. Given a set of mutually exclusive events whose union are the entire sample space $F_1, F_2, \dots, F_n, \mathbb{P}(F_i) > 0 \quad \forall i \in \mathbb{N}$ and such that $E \subset \bigcup_{i=1}^n F_i$

$$\mathbb{P}(E) = \sum_{i=1}^n \mathbb{P}(E|F_i)\mathbb{P}(F_i).$$

1.5 Bayes Theorem / Rule

Bayes rule states that $\mathbb{P}(A|B) = \frac{\mathbb{P}(A)\mathbb{P}(B|A)}{\mathbb{P}(B)}$ for $\mathbb{P}(B) > 0$. This can be extended with the use of the law of total probability to replace the bottom with a sum.

2 Discrete Random Variables and Discrete Distributions

A random variables is a function namely, Let X be a random variable, thus $X : \Omega \implies E$. E is the codomain and defines the set of Values that X can take on. X is assigning numeric output from the outcomes in the sample space. Really it is mapping from the probability space to the space $(E, \mathcal{E})$. Measurability is a property of sets that allows us to take any element in the sigma algebra of E , and if we take the pre-image with respect to X we are guaranteed to map to an element in the sigma algebra. A **probability mass function** is a function that assumes real values in the unit interval to elements in the set E . X is said to be **discrete** if E is a countable set (namely finite or denumerable).

2.1 The Expectation and Variance

We define the expectation for a E -valued random variable X as the quantity

$$\mathbb{E}[X] = \sum_{x_i \in E} p_X(x_i) \cdot x_i. \text{ Define the Variance of } X \text{ as the quantity } \text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

2.2 The Law of the Unconscious Statistician

The Law of the unconscious statistician states that given a function of a random variable $Y = f(X)$ we have that $\mathbb{E}[f(X)] = \sum_{x_i \in E} p_X(x_i) \cdot f(x_i)$. This can be proven considering the preimage of individual y s and manipulating the sum for the expectation. The proof is a little verbose and can be found in Taylors notes.

2.3 The Bernoulli Distribution

A random variable X with parameter p is Bernoulli distributed if $\mathbb{P}(X = 0) = 1 - p, \mathbb{P}(X = 1) = p$. X takes values in the finite set $\{0, 1\}$. Some properties worth noting:

- Bernoulli trials represent trials where the outcome is either success or failure

- $\mathbb{E}[X] = p, \text{Var}(X) = p(1 - p)$
- The Binomial, Geometric, and Poisson are all built upon the Bernoulli Distributed Random Variable

2.4 The Binomial Distribution

Suppose we have n random variables denoted $X_1, X_2, \dots, X_n$ where the random variables are IID (independent identically distributed) Bernoulli with parameter p . Let's say we wanted to consider the probability $\mathbb{P}(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$, because these variables are IID we can decompose this into a product of probabilities. Furthermore, we can define $X = \sum X_i$. X can be at most n and at least 0. It represents the number of success we have in n IID Bernoulli trials. Thus we might be interested in $\mathbb{P}(X = k)$. There are $\binom{n}{k}$ ways to yield k successes in n trials, and since each success and failure is independent we can factor success and failures as follows $p^k(1 - p)^{n-k}$. Thus $\mathbb{P}(X = k) = \binom{n}{k}p^k(1 - p)^{n-k}$. The Binomial Distribution has the following properties:

- $\mathbb{E}[X] = np, \text{Var}(X) = np(1 - p)$
- The Binomial often represents number of success in n trials, it has two parameters n and p , and it is extended to the Poisson Distribution.

2.5 The Geometric Distribution

Given a random variable X that is Geometrically distributed with parameter p , $\mathbb{P}(X = k)$ is the probability that you succeed on the k th trial (this is one definition there are alternative definitions). Namely $\mathbb{P}(X = k) = (1 - p)^{k-1}p$. The Geometric has some interesting properties:

- It is memoryless $\mathbb{P}(X > n + m | X > m) = \mathbb{P}(X > n)$
- $\mathbb{E}[X] = \frac{1-p}{p}, \text{Var}(X) = \frac{1-p}{p^2}$

2.6 The Poisson Distribution

A random variable is Poisson with parameter $\lambda > 0$ if $\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$. A few questions arise while defining this distribution. How does it arise? The poisson represents events with very low frequency, a distribution that is good at approximating the incidence of rare events. Consider a random variable X which is binomially distributed with parameters $(\frac{\lambda}{n}, n)$. I claim that as n goes to infinity the Binomial distribution will approach the

Poisson given the definition of its parameters as above.

$$\begin{aligned}
\lim_{n \rightarrow \infty} \mathbb{P}(X = k) &= \lim_{n \rightarrow \infty} \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\
&= \frac{\lambda^k}{k!} \lim_{n \rightarrow \infty} \frac{n!}{n^k (n-k)!} \left(1 - \frac{\lambda}{n}\right)^{-k} \left(1 - \frac{\lambda}{n}\right)^n \\
&= \frac{\lambda^k}{k!} \frac{n(n-1)(n-2)\dots(n-k+1)}{n^k} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-k} \\
&= e^{-\lambda} \frac{\lambda^k}{k!}
\end{aligned}$$

The mean and variance of the poisson is λ

3 The PGF and Multivariate Distributions

3.1 The Probability Generating Function

We define the probability generating function as $\phi_X(s) = \mathbb{E}[s^X] = \sum_{k=1}^{\infty} \mathbb{P}(X = k) \cdot s^k$. The PGF has some interesting properties. Firstly we find that $\phi_X(0) = \mathbb{P}(X = 0)$, $\phi_X(1) = 1$, $\phi'_X(1) = \mathbb{E}[X]$, $\phi''_X(1) = \mathbb{E}[X^2] - \mathbb{E}[X]$. **Theorem:** Suppose that we have two discrete real valued random variables with values in the non-negative integers, X and Y have the same distribution if and only if their PGFs are the same for all real s. **Theorem:** If a random variable X is a sum of independent random variables the random variable's X PGF is the product of the PGFs of the individual random variables it sums to.

3.2 Multivariate Distributions

Suppose we have n random variables where each random variable X_i takes values in a set E_i . The joint pmf that maps from $E_1 \times E_2 \times \dots \times E_n \implies [0, 1]$ is the probability $\mathbb{P}(X_1 = x_1, X_2 = x_2, X_3 = x_3, \dots, X_n = x_n)$.

3.3 Independence

A set of jointly distributed random variables is said to be independent if the probability of each random variable taking values in some subset of their codomain is the product of the individual probability of a given RV taking values in a subset of the codomain. We say that a set of random variables are independent if and only if the joint pmf can be factored into the product of individual pmfs.

3.4 The conditional Expectation and LOIE

Let X and Y be random variables defined on the same probability space, then the conditional expectation of X given Y = y is equivalent to $\mathbb{E}[X|Y = y] = \sum_{x_i \in E_1} \mathbb{P}(X = x_i|Y =$

$y) \cdot x_i$. When X and Y are independent the conditional expectation is equivalent to the expectation of X . The random variable $\mathbb{E}[X|Y]$ maps from the sample space to the real line, and it is a function of y . Namely $\mathbb{E}[X|Y](w) = \sum_{y \in E_2} \mathbb{E}[X|Y = y] \cdot 1_{\{Y=y\}}(w)$.

Law of the Iterated Expectation:

This law states that the $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X|Y]]$

Proof:

$$\begin{aligned}
 \mathbb{E}[\mathbb{E}[X|Y]] &= \sum_{y_i \in E_2} \mathbb{E}[X|Y = y_i] \mathbb{P}(Y = y_i) \\
 &= \sum_{y_i \in E_2} \sum_{x_j \in E_1} \mathbb{P}(X = x_j | Y = y_i) \cdot x_j \cdot \mathbb{P}(Y = y_i) \\
 &= \sum_{y_i \in E_2} \sum_{x_j \in E_1} \mathbb{P}(Y = y_i | X = x_j) \cdot x_j \cdot \mathbb{P}(X = x_j) \\
 &= \sum_{x_j} \mathbb{P}(X = x_j) \cdot x_j \sum_{y_i} \mathbb{P}(Y = y_i | X = x_j) \\
 &= \mathbb{E}[X]
 \end{aligned}$$

4 Continuous Random Variables and Distributions

4.1 The CDF

Suppose X is a real valued Random Variable, then the CDF of X is the function $F : \mathbb{R} \Rightarrow [0, 1]$ such that $F(x) = \mathbb{P}(X \leq x)$

Properties:

1. Monotonic Increasing $x < y \implies F(x) \leq F(y)$
2. $\lim_{x \rightarrow \infty} F(x) = 1, \lim_{x \rightarrow -\infty} F(x) = 0$
3. Right Continuity, for a sequence $(x_n), n \in \mathbb{N}$ $\lim_{n \rightarrow \infty} F(x_n) = F(x)$, where (x_n) is decreasing
4. For $a, b \in \mathbb{R} \forall_{a < b} \mathbb{P}(a \leq X \leq b) = F(b) - F(a)$

4.2 Continuous Random Variables

A real valued random variable is said to be continuous if there exists a non-negative function $p : \mathbb{R} \Rightarrow [0, \infty)$ such that $\forall_{a < b} F(b) - F(a) = \int_a^b p(x) \cdot dx$. This function p is called the probability density function.

A key property of the PDF is that $\mathbb{P}(-\infty < X < \infty) = \int_{-\infty}^{\infty} p(x) \cdot dx = 1$, and this property allows us to check whether a given function is a continuous probability density function. There is a very close connection between the CDF and the PDF. Recall that

$F(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x p(y) \cdot dy$, Since the right side is differentiable, we know that F is differentiable. Thus the following relation is true $p(x) = F'(x)$, which gives us a simple way to derive the PDF from the CDF.

4.3 The Expectation of Continuous Random Variables

Let X be a continuous random variable with density $p(x)$ then the expectation of X is defined as $\mathbb{E}[X] = \int_{-\infty}^{\infty} p(x) \cdot dx$

4.3.1 LOTUS for Continuous RVs

Lemma: if Y is a non-negative Random Variable with density $p(y)$ then

$$\mathbb{E}[Y] = \int_0^{\infty} \mathbb{P}(Y > y) \cdot dy$$

Proof: $\int_0^{\infty} \mathbb{P}(Y > y) \cdot dy = \int_0^{\infty} (\int_y^{\infty} p(x) \cdot dx) \cdot dy = \int_0^{\infty} (\int_0^x dy) p(x) dx = \int_0^{\infty} p(x) x \cdot dx = \mathbb{E}[Y]$
 $\mathbb{E}[f(X)] = \int_{-\infty}^{\infty} p(x) \cdot f(x) \cdot dx$ Assuming that f is non-negative $\mathbb{E}[f(X)] = \int_0^{\infty} \mathbb{P}(Y > y) dy = \int_0^{\infty} \int_{x:f(x)>y} p(x) \cdot dx = \int_{-\infty}^{\infty} (\int_0^{f(x)} dy) p(x) \cdot dx = \int_{-\infty}^{\infty} p(x) f(x) \cdot dx$. We also know that the expectation of a continuous random variable X is affine, namely $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$.

4.4 The Uniform Distribution

Definition 4.1. A continuous RV has the uniform distribution on the interval (a, b) (denoted $U(a, b)$) if the pdf $p(x)$ of X is

$$p(x) = \begin{cases} \frac{1}{b-a} & x \in (a, b) \\ 0 & \text{otherwise} \end{cases}$$

For any subset of the interval (a, b) $\mathbb{P}(X \in (l, r)) = \frac{r-l}{b-a}$, $(l, r) \subset (a, b)$. Suppose that $X \Rightarrow U(a, b)$ then the n th moment of X is given as $\mathbb{E}[X^n] = \int_a^b x^n (\frac{1}{b-a}) = \frac{1}{b-a} \int_a^b x^n = (\frac{1}{b-a}) \frac{x^{n+1}}{n+1} \Big|_a^b = \frac{1}{n+1} (\frac{b^{n+1}-a^{n+1}}{b-a})$.
 $\mathbb{E}[X] = \frac{1}{2}(a+b)$, $Var(X) = \frac{1}{12}(b-a)^2$ ²

4.5 The Gamma Function and Distribution

Definition 4.2. $\Gamma(a) = \int_0^{\infty} y^{a-1} e^{-y} \cdot dy$ for $a > 0$

Some basic properties $\Gamma(a) = (a-1)\Gamma(a-1)$ for $a > 1$

Proof: $\Gamma(a) = \int_0^{\infty} y^{a-1} e^{-y}$, we approach this integral by parts. Let $u = y^{a-1}$, let $dv = e^{-y}$. Then we know that $du = (a-1)y^{a-2}$, and $v = -e^{-y}$. Integrating by parts we know this integral is further equal to

$$uv - \int v \cdot du = [y^{a-1} \cdot (-e^{-y})] \Big|_0^{\infty} - \int_0^{\infty} (a-1) - e^{-y} (y^{a-2}) = (a-1)\Gamma(a-1) \text{ QED.}$$

²There doesn't exist a random variable distributed on all the reals this follows the fact that the integral of a constant density over an infinite region is infinite

Definition 4.3. The Gamma distribution is defined as follows,

$$\text{Let } p : \mathbb{R} \Rightarrow [0, \infty) \text{ such that } p(x) = \begin{cases} \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

4.6 The Beta Function and Beta Distribution

Definition 4.4. $\beta(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} \cdot dx, (a, b) > 0$

The Beta distribution is defined to be $p(x) = \begin{cases} \frac{1}{\beta(a, b)} x^{a-1} (1-x)^{b-1} & x \in (0, 1) \\ 0 & \text{otherwise} \end{cases}$

An important property $\beta(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$

Proof:

$\Gamma(a)\Gamma(b) = \int_0^\infty u^{a-1} e^{-u} du \int_0^\infty v^{b-1} e^{-v} dv = \int_0^\infty \int_0^\infty u^{a-1} v^{b-1} e^{-(u+v)} du dv$, now to make things easier we will do a change of variables Let $u = st, v = s(1-t)$, then the above integral is further equal to $\int_0^1 \int_0^\infty (st)^{a-1} (s(1-t))^{b-1} e^{-(st+s(1-t))} |J| ds dt = \int_0^1 \int_0^\infty (st)^{a-1} (s(1-t))^{b-1} e^{-(st+s(1-t))} s ds dt = \int_0^1 t^{a-1} (1-t)^{b-1} \int_0^\infty s^{a+b-1} ds dt = \beta(a, b)\Gamma(a+b)$, Thus $\beta(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$

4.7 The Exponential Distribution

Definition 4.5. X is said to have the exponential distribution with rate $\lambda > 0$ if X has the following density.

$$p(x) = \begin{cases} \lambda \cdot e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases} \text{ This distribution is typically used for modeling waiting times, lifespans, etc.}$$

$\mathbb{E}[X] = \int_{-\infty}^\infty xp(x)dx = \int_0^\infty x\lambda e^{-\lambda x} dx$, let $s = \lambda x, ds = \lambda dx$, $= \frac{1}{\lambda} \int_0^\infty e^{-s} s ds = \frac{1}{\lambda} \Gamma(2) = \frac{1}{\lambda}$, the general case, the nth moment is computed $\mathbb{E}[X^n] = \frac{1}{\lambda^n}$

4.7.1 The Exponential distribution is memory-less

Namely, $\mathbb{P}(X > t + s | X > t) = \mathbb{P}(X > s)$

Proof: $\mathbb{P}(X > t + s | X > t) = \frac{\mathbb{P}(X > t+s \cap X > t)}{\mathbb{P}(X > t)} = \frac{\mathbb{P}(X > t+s)}{\mathbb{P}(X > t)}$

$$\mathbb{P}(X > t) = \int_t^\infty \lambda e^{-\lambda u} du = \lambda \int_t^\infty e^{-\lambda u} du = -e^{-\lambda t}, \frac{\mathbb{P}(X > t+s)}{\mathbb{P}(X > t)} = \frac{e^{-\lambda(t+s)}}{e^{-\lambda t}} = e^{-\lambda s} = \mathbb{P}(X > s)$$

Definition 4.6. Suppose that X is a real-valued Random Variable, the Moment generating function of X is the function

$$\phi_X : (-\infty, \infty) \Rightarrow [0, \infty] \text{ defined by } \phi_X(t) = \mathbb{E}[e^{tX}]$$

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³Recall that the PGF is defined as $\phi_X(s) = \mathbb{E}[s^X] = \sum_{k=0}^\infty \mathbb{P}(X = k) \cdot s^k, \forall s \in R$, so the MGF is when $s = (e^t)$

Using the Taylor expansion of e^t , $\mathbb{E}[e^{tX}] = \mathbb{E}[\sum_{n=0}^{\infty} \frac{t^n X^n}{n!}]$, this is not guaranteed to commute in general, but if the moments are finite, we can use the linearity of expectation. $= \sum_{n=0}^{\infty} \mathbb{E}[\frac{t^n X^n}{n!}] = \sum_{n=0}^{\infty} \frac{t^n}{n!} \mathbb{E}[X^n]$.

Properties:

$$\begin{aligned}\phi_X(0) &= \mathbb{E}[e^{tX}] = \mathbb{E}[1] = 1 \\ \frac{d}{dt}\phi_X(0) &= \mathbb{E}[Xe^{tX}]|_{t=0} = \mathbb{E}[X] \\ \frac{d^2}{dt^2}\phi_X(0) &= \mathbb{E}[X^2 e^{tX}]|_{t=0} \\ \phi_X^{(n)}(0) &= \mathbb{E}[X^n]\end{aligned}$$

Suppose that $X_1, X_2, \dots, X_n$ are independent real valued RVs. Let $X = \sum_{i=1}^n X_i$, then $\phi_X(t) = \prod_{i=1}^n \phi_{X_i}(t)$

Proof:

$$\begin{aligned}\mathbb{E}[e^{t(\sum_{i=1}^n X_i)}] \\ &= \mathbb{E}[\prod_{i=1}^n e^{tX_i}] \\ &= \prod_{i=1}^n \mathbb{E}[e^{tX_i}]\end{aligned}$$

Problem: Let X_1, X_2 be real valued random variables such that the following properties are satisfied

- X_1, X_2 are independent
- $X = \frac{X_1 + X_2}{\sqrt{2}}$
- X, X_1, X_2 all have the same distribution
- $\mathbb{E}[X] = 0, Var(X) = 1$

We can start by simplifying the MGF

$$\begin{aligned}
\phi_X(t) &= \mathbb{E}[e^{tX}] = \mathbb{E}[e^{t(\frac{X_1+X_2}{\sqrt{2}})}] \\
&= \mathbb{E}[e^{\frac{t}{\sqrt{2}}X_1}] \cdot \mathbb{E}[e^{\frac{t}{\sqrt{2}}X_2}] \\
\phi_X(t) &= (\phi_X(\frac{t}{\sqrt{2}}))^2 \\
\log(\phi_X(t)) &= \log((\phi_X(\frac{t}{\sqrt{2}}))^2) \\
&= \psi_X(t) = 2\psi_X(\frac{t}{\sqrt{2}}), \psi = PGF(X)
\end{aligned}$$

After deriving the nth derivative, we find that $\psi_X^{(n)}(0) = 2^{1-\frac{n}{2}}\psi_X^{(n)}(0) \Rightarrow 0(n \neq 2)$. The only function that satisfies this equation is $\psi_X(t) = \frac{1}{2}t^2, \phi_X(t) = e^{\frac{t^2}{2}}$

4.8 The Normal Distribution

Definition 4.7. We say that X has the normal distribution with mean μ and variance σ^2 , where the variance is finite, if X has the following density.

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

We can find the moment generating function of the standard normal (parameters mean = 0, variance = 1) as follows

$$\begin{aligned}
\mathbb{E}[e^{tX}] &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)} e^{tx} \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x-t)^2 + \frac{t^2}{2}} dx \\
&= \frac{e^{t^2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{y^2}{2}}, y = x - t \\
&= e^{\frac{t^2}{2}}
\end{aligned}$$

Where does $\frac{1}{\sqrt{2\pi}}$ come from? It is a normalizing constant, that allows the integral of the density over the real line to equal 1. Consider X , suppose that X is a normally distributed (mean μ , variance σ^2). Then the density looks like $p(x) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{x^2}{2\sigma^2}}$, we ignore the μ , since shifting the function over the real line has no effect on the integral. We use u-sub to make this integral more simple, $u = \frac{x}{\sigma}, du = \frac{1}{\sigma}$

5 Advanced Topics

Important Theorem:

Suppose that $X_1, \dots, X_n$ are independent continuous random variables with marginal densities $f_1, \dots, f_n$. Let $h_1, \dots, h_n$ be functions from the reals to the reals then the expectation

of the product of functions applied to the random variables is equivalent to the product of the expectations of the functions applied to the random variables.

5.1 Covariance and Correlation

Define Covariance of two random variables, $Cov(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$, this can be more easily computed as the $\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$.

The following are some key properties that are worth knowing.

$$\begin{aligned} Cov(X, X) &= Var(X) \\ Cov(X, Y) &= Cov(Y, X) \\ Cov(aX + b, cY + d) &= a \cdot c \cdot Cov(X, Y) \\ Cov\left(\sum_{i=1}^n X_i, \sum_{j=1}^m Y_j\right) &= \sum_{i=1}^n \sum_{j=1}^m Cov(X_i, Y_j) \\ Var\left(\sum_{i=1}^n X_i\right) &= 2 \sum_{i < j} Cov(X_i, X_j) + \sum_{i=1}^n Var(X_i) \end{aligned}$$

. Let X, Y be real valued random variables, if the Covariance between these two variables is equal to 0, the variables are said to be **Uncorrelated**. Uncorrelated variables are not necessarily independent, but All independent variables are necessarily uncorrelated. This can be shown formally by computing the Covariance, and the converse can be proven by counter example.

5.2 The Sample Mean, Inequalities

5.3 Sample Mean Definition

Suppose that $X_1, \dots, X_n$ be a sequence of real valued random variables.

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

5.3.1 Proposition: Mean and Variance of the Sample Mean

Suppose $X_1, \dots, X_n$ are IID real valued random variables

With mean $\mu = E[X_i]$, $\sigma^2 = Var(X_i)$. Then $E[\bar{X}_n] = \mu$, $Var(\bar{X}_n) = \frac{\sigma^2}{n}$

Proof:

$$E[\bar{X}_n] = E[\frac{1}{n} \sum_{i=1}^n X_i] = \frac{1}{n} E[\sum_{i=1}^n X_i] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \mu$$

$$Var(\bar{X}_n) = Cov(\bar{X}_n, \bar{X}_n) = \frac{1}{n^2} Cov(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i) = \sum_{i=1}^n \sum_{j=1}^n Cov(X_i, X_j) = \frac{1}{n^2} \sum_{i=1}^n Var(X_i) = \frac{\sigma^2}{n}$$

5.4 Markov's Inequality

Suppose X is a non negative Real Valued Random Variable.

For $t > 0$, $P(X > t) \leq \frac{E[X]}{t}$.

Define the following indicator variable

$$1_{\{X>t\}} = \begin{cases} 1 & \text{if } X > t \\ 0 & \text{if } X \leq t \end{cases} \quad (1)$$

Claim: $t \cdot 1_{\{X>t\}} \leq X$

This is true for all $t > 0$ (Consider cases of the indicator function). In general we know that $X \leq Y \implies E[X] \leq E[Y]$, this further implies $E[t \cdot 1_{\{X>t\}}] \leq E[X]$. We know that the expectation of the indicator variable is the probability of the event that the indicator variable is indicating.

$$tP(X > t) \leq E[X] \implies P(X > t) \leq \frac{E[X]}{t}.$$

5.5 Chebyshev's Inequality

Let X has finite mean and finite variance, let t be a strictly positive variable. $P(|X - \mu| > t) \leq \frac{\sigma^2}{t^2}$.

This proof begins by redefining the inner equality as a new random variable, and squaring Y . Once Y is squared, we can use Markov's Inequality.

5.6 Convergence Methodologies

Suppose that we have a sequence of Real Valued Random Variables that converge in probability to c . If f is a function from the reals to the reals, which is continuous at c , then it can be shown that the application of f onto the sequence of Rv's $f(X_n) \rightarrow f(c)$. The proof is relatively basic, and requires some manipulation and creativity.

5.7 Convergence

Types of Random Variable Convergence

1. Convergence in Probability

$$\forall_{\epsilon > 0} \lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| > \epsilon) = 0 \quad (2)$$

2. Convergence Almost Surely

$$P(\lim_{n \rightarrow \infty} \bar{X}_n = \mu) = 1 \quad (3)$$

3. The previous two conditions require the variables be defined on the same probability space, the next two do not require this condition.

4. Convergence in distribution

$$\forall_{x \in R} (F_x \text{ cont}) \wedge \lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x) \quad (4)$$

This condition is more relevant for continuous random variables, as the CDFs of continuous random variables are necessarily continuous, because they must be differentiable to define the density.

5. Convergence Weakly

$$\forall_f (f \text{ cont}) \wedge \lim_{n \rightarrow \infty} E[f(X_n)] = E[f(x)] \quad (5)$$

There are other forms of convergence like convergence in total variation, but these four are the mainly important notions of convergence.

The Strength of these conditions is as follows

$$CAS \implies CP \implies CD \longleftrightarrow CW$$

5.8 Weak Law of Large Numbers

Suppose that we have a sequence of random variables, IID, with finite mean and variance. Then the sample mean converges in probability to the true mean.

Convergence in Probability:

$$\forall_{\epsilon > 0} \lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| > \epsilon) = 0 \quad (6)$$

Proof:

By using chebyshevs inequality we can say that $P(|\bar{X}_n - \mu| > \epsilon) \leq \frac{Var(\bar{X}_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2}$
 $\lim_{n \rightarrow \infty} \frac{\sigma^2}{n\epsilon^2} = 0$ By squeeze theorem or the property of probabilities this is true. This final quantity we obtained is a confidence level, the smaller it is the more confident we can be that the difference between the sample mean and true mean is very close to 0. Epsilon can be viewed as a sort of confidence level

5.9 Strong Law of Large Numbers

Strong Law of Large Numbers states that if we have a sequence of random variables, IID, with finite mean and variance. Then the sample mean converges almost surely to the true mean. This is a stronger condition, because it can be shown that Convergence almost surely implies convergence in probability.

5.10 The Central Limit Theorem

Suppose that we have a sequence of IID Real Valued Random Variables with finite mean and finite variance (positive). Let $Z_n = \frac{\sum_{i=1}^n (X_i - \mu)}{\sigma\sqrt{n}}$. For every real x we can say that $Z_n \xrightarrow{d} Z$ Where Z is a normally distributed random variable with mean 0 and variance 1.
⁴. Utilizing techniques involving the characteristic equation, which is an imaginary cousin of the MGF it can be shown that this convergence in distribution exists

⁴It can be shown that Z_n is unbounded, namely that its inf is negative infinity and its sup is positive infinity both with probability one.